Practical No. 16: * Write a program to pass function argument into React program and Implement the life cycle of React

□ IX. Conclusion

Answer:
React allows efficient development of web applications by using components. Passing functions as props helps in communication between parent and child components. Lifecycle methods help control component behavior during different stages like mounting, updating, and unmounting. This improves performance and code management.

□ X. Practical Related Questions

Q1. Write a program to pass function argument into React component

program1.js

```
import React, { useState } from "react";

function Program1() {
  const [count, setCount] = useState(0);

  // Function with argument
  const addValue = (value) => {
    setCount(count + value);
  };

  return (
    <div style={{ textAlign: "center" }}>
      <h2>Count: {count}</h2>

      {/* Passing function with argument */}
      <Child add={addValue} />
    </div>
  );
}

function Child(props) {
  return (
    <div>
      <button onClick={() => props.add(1)}>+1</button>
      <button onClick={() => props.add(5)}>+5</button>
      <button onClick={() => props.add(10)}>+10</button>
    </div>
  );
}

export default Program1;
```

Q2. Implement the Life Cycle of React

program2.js

```
import React, { Component } from "react";

class Program2 extends Component {
  constructor() {
    super();
    this.state = { count: 0 };
    console.log("Constructor called");
  }

  componentDidMount() {
    console.log("Component Mounted");
  }

  componentDidUpdate() {
    console.log("Component Updated");
  }

  componentWillUnmount() {
    console.log("Component Will Unmount");
  }

  render() {
    return (
      <div style={{ textAlign: "center" }}>
        <h2>Count: {this.state.count}</h2>
        <button onClick={() => this.setState({ count: this.state.count + 1
})}>
          Update
        </button>
      </div>
    );
  }
}

export default Program2;
```

App.js

☞ **Run only one program at a time** by commenting/uncommenting.

```
import React from "react";
import Program1 from "./program1";
import Program2 from "./program2";

function App() {
  return (
    <div>
      {/* Run Program 1 */}
      {/* <Program1 /> */}

      {/* Run Program 2 */}
      <Program2 />
    </div>
  );
}

export default App;
```

□ XI. Exercise

Q1. Why is an arrow function used inside the onClick event?

Answer:
Arrow functions are used because they automatically bind `this` to the component. They also allow passing a function without executing it immediately. This ensures correct behavior of event handling.

Q2. What is the difference between Mounting and Updating phases?

Answer:

- **Mounting Phase:**
  When the component is created and added to the DOM.
  Methods used: `constructor()`, `componentDidMount()`

- **Updating Phase:**
  When the component is re-rendered due to changes in state or props.
  Method used: `componentDidUpdate()`